

Software Requirements Specification

For

GharBazar Web Application

1) Introduction

1.1 Purpose

The purpose of this Software Requirement Specification is to define the requirements for GharBazar. This document has outlined the web-app functionalities and non-functionalities requirements, design, system dependencies, and performance expectations

1.2 Document Conventions

- **Bold** text is used to show high-priority items or definite UI elements.
- Tags that are labeled with [MVP] will be compulsory in the first release.

1.3 Intended Audience and Reading Suggestions

- **Developers:** Section 4 (System Features): Section 3 (External Interfaces).
- **Stakeholders:** See Section 1 and 2 and make sure that it achieves the objective of removing the traditional process of property search in Nepal.

1.4 Product Scope

GharBazar is an online platform that makes property transactions in Nepal easier through providing property management, search/filter, ratings/reviews, favorites, and direct communications. It aims at property owners, buyers, renters, and admins and tackles such classic problems as the lack of accessibility and the lack of trust with the help of a responsive web application.

2) Overall Description

2.1 Product Perspective

GharBazar is an independent Web site, designed using React.JS on the front end and ASP.NET core on the back end, connected to the MySQL database. It has the need of the internet and it incorporates email notification services.

2.2 Product Functions

- **Normal Users:** Property browsing, advanced search/filter.
- **Registered Users:** add/edit properties (owners), rate/review, save favorites, manage profiles.
- **Admins:** Control board, user/property control.

2.3 User Classes and Characteristics

- **Guests:** Casual visitors that want to know more about the property.
- **Buyers/Renters:** Fast search and favorites that are frequently used by buyers
- **Property Owners:** Edit / upload property listings.
- **Admins:** Manage control and monitoring of the system.

2.4 Operating Environment

Client Side:

- Modern Desktop or Laptop with minimum 4GB RAM(8GB Recommended)
- Mobile Devices with minimum 2GB RAM
- Minimum screen resolution 1280 x 720 p for desktop
- Stable Internet Connection

Server Side:

- Web Server With minimum 8GB RAM (16GB recommended)
- Minimum 50GB SSD storage for database and media files
- Multi core processors
- Backup storage for database backup.

3) External Interface Requirements

3.1 User Interface

User interface is to be simple, contemporary real estate-based and responsive design based on Tailwind CSS/Bootstrap.

- **Landing Page:** Prominent search bar, Featured Properties
- **Property Details Page:** Large property pictures, description, location, and obvious button of Contact Owner.
- **Search/Filter Panel:** Visual filters of location, price range, type of property (house/apartment/land), bedrooms, facilities.
- **User Dashboard:** Basic list of "My Favorites" and "My Properties" (owners) with an affiliate button that saves or un-saves the entries.
- **Admin Dashboard:** tabular user, property, review and bulk action and analytics chart management.

3.2 Hardware Interface:

- **Server:** Standard cloud compute instances (8GB+ RAM, 50GB SSD or larger) are required.
- **Client:** Web applications are sensitive to touch (mobile) and mouse (desktop); property photos can be uploaded.

4) System Features

4.1 Authentication of users (High, MVP)

- REQ-1.1 Create an account/register/account creation/registration/conscription.
- REQ-1.2 Login via credentials
- REQ-1.3 Password reset via email
- REQ-1.4 Role based access (owner/buyer/admin)

4.2 Property Management (High, MVP)

- REQ-2.1 Owners CRUD properties (add/edit/ delete/ update)
- REQ-2.2 Image uploads (max 5, 10MB each)

4.3 Search & Filter (High, MVP)

- REQ-3.1 Filters on key words/locations/price/type.

- REQ-3.2 Auto-suggest on 3-character basis.

4.4 Ratings & Reviews (Medium)

- REQ-4.1 5-star rating + text feedback (authenticated only)
- REQ-4.2 Admin publish mod.

4.5 Favourites & Profile (Medium)

- REQ-5.1 View/ save favorite properties.
- REQ-5.2 Profile: see/update/add picture.

5) Additional Nonfunctional Requirements.

5.1 Performance Requirements

- **Load Time:** The home page should also load in 1.5 seconds using regular 4G connection.
- **Capacity:** System should be able to serve 2,000 parallel users and 300 parallel searches of property every minute.
- **Scalability:** Backend should be able to add server instances automatically in case CPU usage goes beyond 70%. Security Requirements

5.2 Safety Requirements

- **Data Backups:** Automated property database backups after every 6 hours.
- **Failover:** read-only replica comes online within 60 seconds in case of failure of primary database to do property browsing.

5.3 Security Requirements

- **Encryption:** passwords has been hashed by BCrypt.
- **Sanitization:** Any user input is sanitized to SQL Injection and XSS attacks.
- **Access Control:** admin routes can only be accessed by users with role of ADMIN.

6) Other Requirements

6.1 Assumptions and Dependencies.

6.1.1 Assumptions:

- User must have a good internet connection and compatible devices
- Device must meet the minimum requirements

6.1.2 Dependencies:

- This app will require a backend server to store and manage user data.

- This app will be developed using react and .net